

Synthetic Citation Practices in Automated Bibliography Extraction

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The measurement of scholarly impact has long depended on the careful parsing of printed bibliographies. Early approaches relied on manual transcription, a process both laborious and prone to systematic error. Subsequent work demonstrated that citation contexts carry substantial rhetorical information about the citing work¹. A recurring finding is that reference lists exhibit remarkable regularity within a journal but striking diversity across disciplines. Extraction pipelines must therefore tolerate variation in punctuation, ordering, and typographic convention. Several studies have shown that author name disambiguation remains the principal bottleneck in large scale indexing². The problem is compounded by initials, diacritics, and institutional authors that defy naive parsing rules. Optical character recognition introduces further noise, particularly in documents scanned from bound volumes. Typographic ligatures and discretionary hyphens are frequent sources of spurious tokens in extracted text³. Cross-column reading order errors account for a significant fraction of failures in two column layouts. Page furniture such as running heads and folios is routinely mistaken for bibliographic content. Robust segmenters exploit whitespace geometry rather than relying on lexical cues alone⁴. Supervised sequence labelling has emerged as the dominant paradigm for reference string parsing. Feature based models remain competitive when training data is scarce or domain shifted. The availability of style aware synthetic corpora has materially improved evaluation practice⁵. Ground truth construction, however, continues to demand painstaking manual verification. Digital object identifiers provide a strong anchor when present, yet coverage is far from uniform. Older literature is disproportionately affected by missing or malformed persistent identifiers⁶. Conference proceedings and theses occupy a grey zone in many citation style definitions. Preprint servers have introduced new conventions that established styles accommodate only awkwardly. The interplay between volume, issue, and pagination fields is a perennial source of ambiguity⁷. Renderer differences between processor implementations further complicate cross platform comparison. Evaluation protocols should therefore report style level metrics rather than a single aggregate score. Error analysis reveals that truncation of long author lists is mishandled by a surprising number of systems⁸. The et alii device in particular interacts badly with naive regular expression approaches. Title casing conventions differ between styles and must be normalised before string comparison. Accented characters survive some extraction pipelines and are silently normalised away by others⁹. A conservative normalisation strategy preserves the grapheme inventory of the source document. Downstream applications such as recommender systems are sensitive to even small error rates. Bibliographic coupling analyses assume near perfect recall of the underlying reference lists¹⁰. The methods section of this paper details a fully synthetic yet typographically faithful corpus. Each document was typeset programmatically under controlled layout perturbations. Imperfections were injected only where they plausibly arise in real production scanning workflows¹¹. The corpus spans author date and numeric citation traditions across ten widely used styles. All items are fictional and any resemblance to real publications is entirely coincidental. We release the corpus together with the generation scripts to encourage replication and extension¹².

Bibliography

- Almeida, Rui J. *Principles of Computational Hydrology: Models, Methods, and Uncertainty*. 2nd ed. Oxford University Press, 2001.
- Anderson, Ruth M., and João P. Silva. "Civic Participation and Trust in Municipal Government: A Longitudinal Study of Twelve Mid-Sized Cities." *Urban Affairs Review* 48, no. 5 (2012): 655–84.
- Ashworth, Priya, and Finn Lindgren. "Variational Inference for Sparse Gaussian Process Modulated Point Processes." In *arXiv Preprint*. Preprint, 2008. 0806.2143. arXiv. <https://arxiv.org/abs/0806.2143>.
- Barakat, Layla, and Erik Johansson. "Soil Carbon Turnover under Contrasting Tillage Regimes in Semi-Arid Cropland." *Soil Biology and Biochemistry* 38, no. 9 (2006): 2451–63.
- Bergeron, Chloé. "Machine-Learned Potentials for Reactive Interfaces: Promises and Pitfalls." In *Frontiers in Atomistic Simulation: From Electrons to Engineering*, edited by Ari Goldman and Mia Stevenson. Springer, 2014. https://doi.org/10.1007/978-3-319-05274-4_9.
- Devereux, Aoife, and Thandiwe Mbeki. "Federated Fine-Tuning of Compact Language Models for Low-Resource Translation." *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics* (Stroudsburg, PA), 2023, 7821–36. <https://doi.org/10.18653/v1/2023.acl-long.433>.
- Fernández-Ríos, María José, Piotr Łukaszewski, and Adaeze Onyema. "Catalytic Upcycling of Polyethylene Waste into Liquid Alkanes over Ruthenium Nanoparticles." *ACS Catalysis* 3, no. 10 (2013): 2287–96. <https://doi.org/10.1021/cr400000a>.

doi.org/10.1021/cs400517k.

- Fitzgerald, Maeve, Carlos Duarte, and Soo-Jin Hwang. "Low-Power Mesh Routing for Structural Health Monitoring of Suspension Bridges." *Proceedings of the 9th ACM Conference on Embedded Networked Sensor Systems* (New York), 2004, 312–25. <https://doi.org/10.1145/1021446.1021478>.
- Grant, Olivia. "Open Data Policies for National Meteorological Services: A Comparative Review." Global Data Commons Initiative, March 14, 2009. <https://www.datacommons-initiative.org/reports/met-services-review>.
- Hartwell, Diane C., and Patrick L. Ng. "Adaptive Filtering of Seismic Reflection Profiles Using Wavelet Packet Decomposition." *Journal of Applied Geophysics* 41, no. 2 (1998): 113–29. [https://doi.org/10.1016/S0926-9851\(98\)00021-4](https://doi.org/10.1016/S0926-9851(98)00021-4).
- Haugen, Ingrid B., and Rodrigo Castellanos. "Migration Timing of Arctic Shorebirds Responds Unevenly to Spring Warming." *Global Change Biology* 21, no. 8 (2015): 3011–23. <https://doi.org/10.1111/gcb.12904>.
- Holmberg, Astrid, Tomás Reyes-García, and Harpreet Singh. "Single-Cell Transcriptomic Atlas of Regenerating Zebrafish Myocardium at Unprecedented Resolution." *Nature Communications* 16 (2025): 1187–203. <https://doi.org/10.1038/s41467-025-56123-x>.
- International Consortium for Coastal Resilience. "Guidelines for Community-Led Coastal Retreat: Synthesis of the Five-Country Pilot Programme." *Climate Risk Management* 6 (2010): 1–18. <https://doi.org/10.1016/j.crm.2010.03.001>.
- Kaminski, Yara, Samuel T. Abebe, and Élodie Fournier. "Edge Computing Architectures for Real-Time Wildfire Detection from Drone Imagery." *Remote Sensing* 13, no. 14 (2021): 2741–60. <https://doi.org/10.3390/rs13142741>.
- Kjærsgaard, Søren M., André Baudouin, and Chinwe Okafor. "Nitrogen Retention in Restored Wetlands of the Kjærsgaard River Basin." *Ecological Engineering* 13, nos. 1–4 (1999): 201–17. [https://doi.org/10.1016/S0925-8574\(99\)00012-7](https://doi.org/10.1016/S0925-8574(99)00012-7).
- Martinez-Campos, Lucia, Sarah J. Whitfield, and Rémi Dubois. "Phonological Priming Effects in Bilingual Lexical Access: Evidence from Eye Tracking." *Journal of Memory and Language* 57, no. 3 (2007): 389–411. <https://doi.org/10.1016/j.jml.2007.04.002>.
- Moreau, Isabelle, and Aleksei Petrov. "Institutional Barriers to Water-Sharing Agreements in Transboundary Basins." In *Governance of Shared Water Resources: Comparative Perspectives*, edited by Beatriz Salazar and Takeshi Kimura. Edward Elgar, 2003.
- Novak, Katarina. "Optical Trapping Dynamics of Colloidal Particles near Dielectric Interfaces." PhD thesis, Delft University of Technology, 2005.
- Okonkwo, Ngozi, and Lars H. Petersen. *The Economics of Urban Green Space: Valuation Methods and Policy Design*. Routledge, 2016.
- Ostrowski, Magdalena, Wei Chen, Funmilayo Adeyemi, et al. "Genome-Wide Association Mapping of Drought Tolerance Loci in Spring Wheat." *Theoretical and Applied Genetics* 104, no. 6 (2002): 987–1001. <https://doi.org/10.1007/s00122-002-0914-6>.
- Quispe-Mamani, Rosa E., and Brendan Callaghan. "Language Revitalization through Community Radio: Evidence from Three Andean Regions." *International Journal of the Sociology of Language* 2019, no. 256 (2019): 55–79. <https://doi.org/10.1515/ijsl-2018-2010>.
- Sharifi, Leila, Stefan Novaković, and Ji-Hye Park. "Deep Convolutional Networks for Automated Detection of Microaneurysms in Retinal Fundus Images." *IEEE Transactions on Medical Imaging* 36, no. 7 (2017): 1490–501. <https://doi.org/10.1109/TMI.2017.2680942>.
- Vasquez, Elena R., and Hiroshi Takeda. "A Distributed Algorithm for Clock Synchronization in Fault-Tolerant Sensor Networks." *IEEE Transactions on Parallel and Distributed Systems* 11, no. 7 (2000): 689–703. <https://doi.org/10.1109/71.877736>.
- Zhao, Mei-Lin, and Tomasz Kowalczyk. "Thermochromic Vanadium Dioxide Thin Films for Passive Radiative Switching." *Applied Physics Letters* 98, no. 14 (2011): 141905–8. <https://doi.org/10.1063/1.3574012>.